Expert Knowledge in Sleep Medicine with Uncertainty through Quantum Fuzzy Inference Systems

Samuel Magaz-Romero Eduardo Mosqueira-Rey Diego Alvarez-Estevez
Vicente Moret-Bonillo

Universidade da Coruña (CITIC), Campus de Elviña, 15071 A Coruña, España

Quantum Fuzzy Inference System is a framework that represents **uncertain knowledge through quantum states**, exploiting quantum superposition as a natural mechanism for encoding fuzzy information, and implements the inference process with quantum operators. We have applied it to the **detection of apneic events in sleep medicine**. The quantum implementation produces inference outcomes that are performance-wise on par with the corresponding classical fuzzy inference system, with a **similarity of 99.97%.**

Motivation

Methods of **classical Artificial Intelligence (AI)** provide the means to **explain their outputs** thanks to their foundation on knowledge engineering, which **current Machine Learning (ML) methods lack**. While ML has gain popularity in the past years, it is of interest to **revisit classical AI through the lens of Quantum Computing**, with its **probabilistic nature being a match for uncertain information**, and exploring its potential advantages.

Background

Fuzzy Inference Systems (FIS) are a realization of an inference system through the means of **fuzzy logic** (Fig. 1). Using the knowledge base from a domain, the inferential process is structured as follows:

- **Fuzzification**: the values of the domain are passed through membership functions to obtain the corresponding fuzzy values

$$\mu_{X^i}(x) : [min_x, max_x] \to [0, 1]$$

- **Inference engine**: a set of rules is applied, according to the activated values of their fuzzy inputs, to obtain fuzzy outputs

$$\text{IF } (x_0 \text{ is } X_0^i) \wedge (x_1 \text{ is } X_1^j) \wedge \cdots \vee (x_n \text{ is } X_n^k) \text{ THEN } (y \text{ is } Y^y) = w_i$$

- **Defuzzification**: the fuzzy outputs are aggregated to obtain a value in the range of the problem's domain, according to constants associated to the outputs

$$y = \sum_{i=0}^{N-1} w_i z_i \Big/ \sum_{i=0}^{N-1} w_i$$

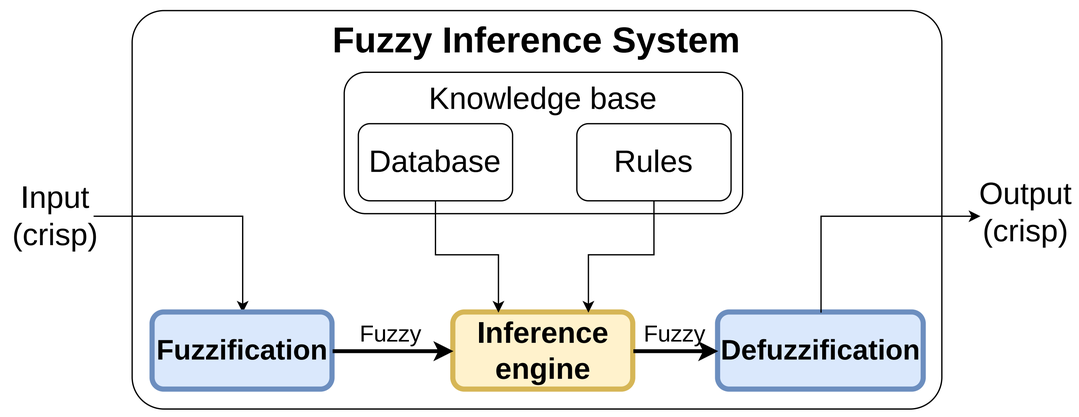


Figure 1. FIS' structure diagram.

Proposal

Quantum Fuzzy Inference Systems (QFIS) are a new implementation of FIS which have the goal of defining the process of **fuzzy inference via quantum computation** (Fig. 2). The fuzzification and defuzzification remain classical, and the inference engine is defined as follows:

- **Quantum state**: the fuzzy values of each input is encoded in the amplitudes of a qubit register

$$Init_X(x) \left|0^n\right\rangle = \left|\psi_x\right\rangle = \sum_{i=0}^{n-1} \sqrt{\mu_{X^i}(x)} \left|i\right\rangle + \sum_{j=n}^{2^q-1} 0 \left|j\right\rangle$$

  The output is initialized to an equal superposition of its states, with an extra qubit to mark the activation of each possible outcome

$$\left|\psi_y\right\rangle = \left|y_{ex}\right\rangle \otimes \left(\frac{1}{\sqrt{n}} \sum_{i=0}^{n-1} \left|i\right\rangle + \sum_{j=n}^{2^q-1} 0 \left|j\right\rangle\right).$$

- **Quantum circuit**: the rules for the inference are implemented through quantum operators controlled by the quantum registers of the inputs, applying an X operator to the extra qubit of the output register, indicating its action

$$RN(\left|\psi_{x_0}\right\rangle \otimes \left|\psi_{x_1}\right\rangle \otimes \ldots \left|\psi_y\right\rangle) = \left|\psi_{x_0}\right\rangle \otimes \left|\psi_{x_1}\right\rangle \otimes \ldots \left|\psi'_y\right\rangle$$

- **Measurement**: the qubit register of the output is measured, and the amplitudes of its states are passed to the defuzzification process to obtain the final value

$$y = \left\langle\psi_y\right| M_y \left|\psi_y\right\rangle$$

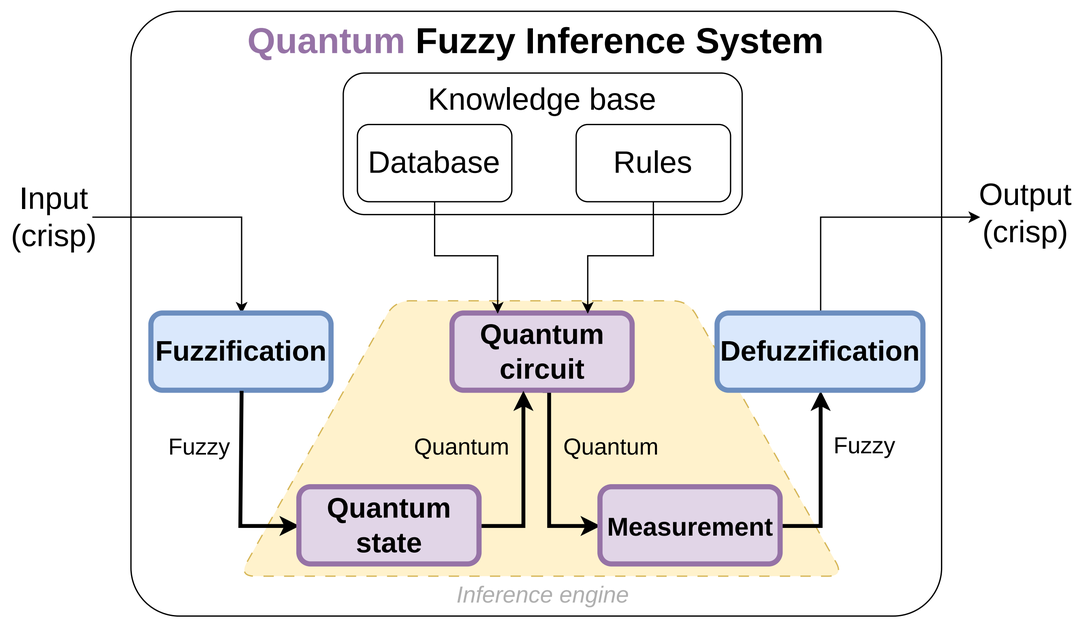


Figure 2. QFIS' structure diagram.

Application

Both FIS and QFIS have been applied to solve the problem of the detection of apneic events in sleep medicine. In this case, a single inferential unit has been implemented, which takes as input single points of two signals: the reduction in saturation of oxygen (SaO_2Red) and the duration of this reduction ($SaO_2RedDur$). The output is the probability of presence of an apneic event ($Event$).

The classical version has been implemented with MATLAB's Fuzzy Logic Designer, which provides the required functionalities for this task. The quantum version has been implemented with IBM's Qiskit, resulting in the quantum circuit of Fig. 3.

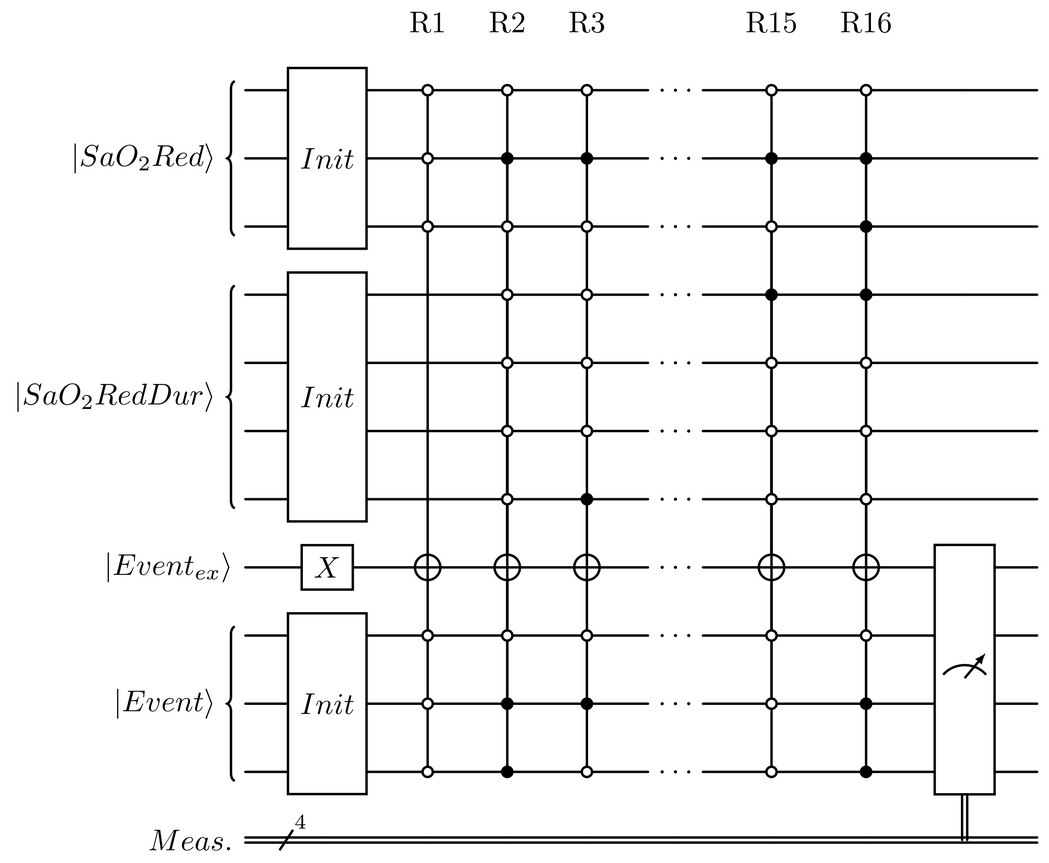


Figure 3. Quantum circuit for the QFIS inference engine.

To test each implementation, an experiment has been conducted, in which each input's range is thoroughly explored, generating the control surfaces for the systems. The results present a geometric similarity of 99.97% (Fig. 4), proving the quantum version as a valid replacement for the classical one.

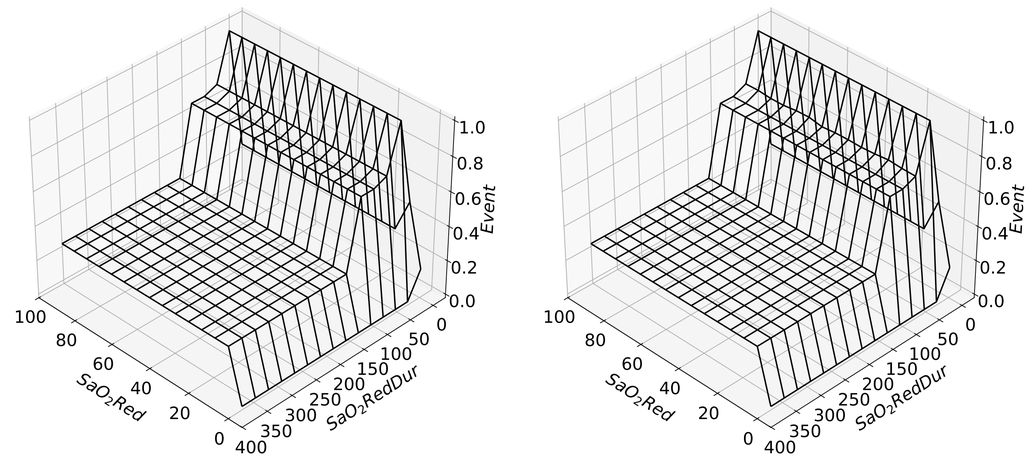


Figure 4. Results for FIS (left) and QFIS (right) applications.

Conclusions

- QFIS is a well defined framework for the development of explainable and uncertainty-aware **Quantum Artificial Intelligence systems**.
- The framework has the potential to support decision-making across domains requiring **expert knowledge, uncertainty, transparent reasoning, and complex inference**.
- Future work will focus on applying quantum techniques focused on **optimization** and execution under **noisy environments**.

Acknowledgements

This study has been supported by project PID2023-147422OB-I00 with funding from grant MCIU/AEI/10.13039/501100011033 and the European FEDER program, and projects ED431C 2026/38 and ED431F 2025/35 funded by Xunta de Galicia. Authors wish to acknowledge the support received from Centro de Investigación de Galicia "CITIC", center accredited for excellence within the Galician University System and a member of the CIGUS Network, which receives subsidies from the Department of Education, Science, Universities, and Vocational Training of the Xunta de Galicia, and it is co-financed by the EU through the FEDER Galicia 2021-27 operational program ED431G 2023/01. DAE has received support from RYC2022-038121-I, funded by MCIN/AEI/10.13039/501100011033 and European Social Fund Plus (ESF+).

Scan this poster!

Corresponding author: s.magazr@udc.es